CHAPTER 13

Science and democracy

A complex relationship

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The article analyses the relationship between science and democracy on basis of a very specific point of view. Three arguments are put forward to defend that. In its historical origin, in its linguistic roots and in its institutional basis, science is (and continues to be) a democratic endeavor. The article very briefly discusses the decisive changings in the relation between science and political, economic and military power that took place during the XX century and stresses the cognitive relevance of a set of universal institutions which, all along the history of science, provide the conditions of possibility for science to exist and develop. Finally, even if today those universal institutions face big transformations, the possibility for science to remain a free democratic endeavor is questioned and argued.

Keywords: philosophy of science, science, democracy, cognitive universal institutions, historical *a priori*, big science, technoscience, citizen science

To think out the relationship between science and democracy is an interdisciplinary task demanding the contribution of a large scope of disciplines: political sciences, philosophy, history, science policy, economics, semiology, sociology, environmental studies, etc.

However, the aim of this article is just to make some suggestions for the analysis of that relationship from the point of a very peculiar approach in the context of philosophy of science. The aim is not to question the similarities of science and democracy, as for instance, to look for what the structural regime of science has inherited from democratic processes or how much the usual qualities of the scientific *habitus* are replicas of the democratic virtues. Nor it is the aim of this article to enter in the discussion of the broad and more programmatic topics of the so-called "politics of science" covering the features concerned with the articulation between

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knowledge production and social impact of science, such as state and private funding, expertise, equipment, research policies, etc. The aim is to focus on the internal structural regime on basis of which science and democracy are grounded.

1. I propose to star on basis of the working hypothesis according to which science is, in its origin, in its root and in its basis a democratic endeavor. Of course, the demonstration of this thesis would claim for great developments, impossible to provide in the context of an article. I will thus reduce the task to just three very brief appointments regarding three levels of analysis: historical, linguistic and transcendental.

In a first (historical) level, science is a democratic endeavor because science and democracy were born together. They had the same origin, that is, the emergence of science largely coincides with the emergence of democracy. In fact, democracy was a Greek invention. Before Greek classical civilization, power was gained by inheritance or by conquest, never before it had been decided by the vote of the citizens. Science was also a Greek invention. Before, important sum of wisdom, huge amount of traditional knowledge, verbally transmitted from the old to the new generations, quantity of empirical results and useful recipes were constituted (Egypt), but only with the Greeks science got the necessary theoretical configuration and demonstrative structure. Now, what it is marvelous to recognize is that the emergence of science largely coincides with the emergence of democracy. Science was born in the Greek democratic *polis*. And – I would add – it could not have been otherwise.

In a second (linguistic) level, science is a democratic endeavor because science and democracy have a common intimate discursive structure, that is, science and democracy share the same discursive requirements at their root. In opposition to the narrative discourse grounded in the past, locally determined, having as its only means of transmission the oral word and the paradigmatic figure of the *oldest*, it emerged in the Greek democratic city a new kind of discourse. A discourse not turned to the past (tradition) but facing what it is, now, in the present (science). A (rational) discourse able to establish and observe a set of normative (logical) rules. The participants were able to get a tacit agreement based in the respect for a set of norms which enable them to coordinate their views, to rule their acts, to validate their statements and to promote the inter-subjective acknowledgement of truth. A demonstrative discourse proceeding by a chain of reasons which any man (even a slave) may approve, share and refute.

What I mean is that science has incorporated the logical procedures emerging in the democratic city. Science has integrated the dialogical practices developed in the city (controversy, rational arguing). Science has absorbed the discussion practices and the legitimating strategies put forward by the wide educated and curious audiences that inhabited the democratic city. It is enough to read one of Plato's dialogues for understanding in which way Greek democratic city is the place, as Stengers says, of "simultaneous emergence of a science of the politics and of a politics of science, a place where it becomes explicit the double problem of the legitimacy of power and of the legitimacy of knowledge" (Stengers, 1998, p. 72).

See the exemplary case of Plato's *Protagoras.* In a lively theatrical way, Plato's *Protagoras* is a fantastic point of observation of the discursive practices in use in V century Greek cities. The dialogue exemplifies and exhibits diverse kinds of arguments. It clarifies the nature of the discursive mechanisms responsible for the invention of the occidental rationality, its formal determinations, its rules and procedures. It allows us to observe the deep changes in human societies where dialogue has just been invented – a technology that is going to change the world, giving rise to the discursive rationality in which science and philosophy have their roots.

The dialectical intensity, the intellectual debate put forward in that memorable meeting at Calias' house points indeed to a series of novelties. The first novelty concerns **equality**, the fact that we are face to a language interchange established among equals, members of a democratic city, linked among them by a similar will of knowledge.[1] The second novelty is the **refusal of authority**, a feature which clearly shows a world in which the spoken word of the older loses the unidirectional road it had before. Now, above the truth, any authority of the older is from now on respected (Socrates is younger than Protagoras). What matters in the democratic city is the capacity of each one (be it a newer) to define with accuracy the concepts under analysis and to build with them a well justified chain of reasons. The third novelty relates **collectiveness and universality**,[2] that is, the fact that, since its very beginning, science is constituted as a collective task, an endeavor which needs to be cooperatively researched. That is why Socrates, in a crucial moment, quotes Homer (*Iliad* X 225): "When too man walk on together, one may see before the other".[3] Science ought to be a largely open joint activity. Scientific truth only makes sense if it belongs to us all. The fourth novelty is **dialectics**, showing how much science is made of a new kind of discursive mechanisms, of new formal determinations, rules and procedures, namely discussion, controversy, argument. Plato's *Protagoras* is the moment in which the very nature of that discursive novelty is questioned and thematized. As Socrates stresses, what matters is not

1. What is not so usual in Plato. See, for instance, *Timeus*' long monologs by Critias, Timeus, Hermocrates and Socrates.

2. Collectiveness and, in fact, universality. However, while collectiveness is a necessary procedure, universality is an endeavor. Science is universal insofar as it aims at universality.

3. Plato, *Protagoras*, 348d.

to seduce with long, beautiful (mythic, poetic) discourses but to formulate short, precise arguments whose diverse kinds Plato exemplifies and exhibits. The issue is not to persuade, to convince, but to make it clear, to explain, to open the way for universal understanding. The sixth novelty concerns **freedom of thought**. In Calias' house there is no restriction or imposition concerning the subjects to be discussed. People decide upon the questions to be pursued according to the conceptual needs of the very investigation. It is precisely because science supposes freedom in the development of its research that it does not work well with anti-democratic regimes. See the exemplary case of Galileo, that foremost sun of the city (Pisa) or Lysenko, or Marr.

But, in my point of view, there is a third (transcendental) level that must be considered. Science is a democratic endeavor because the (institutional) conditions of possibility for science to exist and develop are provided by the democratic city. In other words, it is the democratic city that provides the main institutional basis which make possible the future continuity and advances of scientific knowledge. At a certain moment (precisely that of the invention of democracy) the community of humans organized in a city gave rise, from its own interior and inner forces, to the cognitive configurations that allow the existence of science and make possible its development. From that moment on, a set of institutional practices and material means are prepared to guarantee the new forms of production, conservation, transmission and legitimation of scientific knowledge. Those cognitive configurations are the *Scientific Community* ("République des Savants"), the *School* (namely, the *University*), the *Library* and the *Encyclopedia*,[4] that is, a set of universal institutions, concrete practices and structural procedures pointed out for preserving and organizing knowledge already constituted, for promoting disciplinary coordination and for creating the necessary conditions for future development of scientific knowledge.

> How could science exist without a community of researchers passionately and in full time regime dedicated to the search of truth, free men (not obliged to hard, manual labor) who could entirely devote themselves to contemplation, that is, theory workers. In the Greek democratic city, for the first time appeared these professionals of theory. They were not only allowed, but accepted, stimulated and reimbursed (see Protagoras, the best payed of all sophists). Their work was recognized, admired and praised (the condemnation of Socrates being a mistake and a crime that Plato never accepted and never forgave).

4. Also the Museum (that will not be mentioned here as it implied further developments), while a depurated and typified projection of the world's entities which science aims to understand and which are presented at the museum free of redundancies, is a transcendental cognitive configuration (cf. Pombo, 2011).

> How could science exist without School (University), an institution which offers the new generations the possibility of acquiring the patrimony of knowledge conquered by earlier generations, thus preparing them to continue the adventure of knowledge? See Plato's Academia, Aristotle's Lyceum, or Alexandria's Museion. Those institutions offered the necessary conditions for the constitution of new kinds of knowledge determined by a set of (logical, rational) rules to which both pupil and teacher were submitted, mainly mathematics (precisely that which may better be learned and taught).[5]
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> How would it be possible for science to exist without the invention of Library in whose armoires were ranged the rolls of papyri, the codices, the incunabula, the manuscripts, the letters, the treatises, the books (300.000 in Pergamos's, 700.000 in Alexandria's Library), today, the journals and revues, the dissertations, the papers, the articles and the proceedings, the press and digital documents of all sorts (nowadays, mostly electronic) in which knowledge is condensed, materialized and offered to future generations?[6]
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> And how could Library function without the Encyclopedia as knowledge articulatory device? How was (is) it possible to go deep in knowledge without a map, a cartography, an imagetic horizon of the actual articulation of knowledge domains, a "système des savoirs", as Diderot used to say.

What I mean is that, since its beginnings and throughout its entire historical development, science relies on these universal institutions. Each advancement in knowledge production is prepared by the functioning of these cognitive configurations. They constitute the institutional shadow of science, their material condition of possibility, their historical *a priori* (to put it in Foucauldian terms),[7] simultaneously necessary and tangible, universal and historical.

What I mean is that there will be no science if the Greeks have not invented, in parallel with science, together with it, the scientific community, the school, the library and the encyclopedia.[8] What I mean is that, since the emergence of science and all along its history, those cognitive institutions express, materialize and guarantee the existence of scientific activity, its progress and its democratic nature.

5. From the Greek word *mathemata* (what is taught), *mathematein* (to teach), *mathesis* (what may be taught). For a celebrated exploration of these etymologies, cf. Martin Heidegger (1962, pp. 75–82).

6. We are of course here very close to Popper's 3rd world (Popper, 1972).

7. The expression appears in the "Preface" to *Les Mots et les Choses* (1966), when Foucault defines the analytic work of his book as a quest for the "historical *a priori*" upon which could sciences be constituted.

8. For further developments in the demonstration of precisely this thesis, cf. Pombo (2011).

2. Let us now leave the Greeks behind. They created a splendorous world. But the return is impossible. We must go on.

Since the scientific revolution, it is fair to say that science has always been at the service of the city, even when the city was not democratic. And most of the time, it was not. The case of Giordano Bruno is the extreme example of what could happen when science dare not to obey the well-established social beliefs. Since the scientific revolution it is also fair to say that science had the aim – or, at least, the effect – of contributing for the development of human communities, for the resolution of their problems and for answering their explicative needs. The case of Bacon is eloquent. In the early XVII century, Bacon glorified the enormous services science was prepared to offer the city. As he writes: "the empire of man over things depends wholly on the arts and sciences. For we cannot command nature except by obeying her laws" (Bacon, 1620, p. 129). Later, in the XVIII century, Diderot's *Encyclopédie des Science, Arts et Métiers* (1751–1772) is guided by the enlightened project of showing the universal usefulness of science and its vast applications in the industrial revolution that was emerging in that moment. That is why, according to Diderot, the *Encyclopédie* is "not a state's book but a people's book" (Diderot, 1994, p. 428), able "to gather the knowledge scattered over the surface of the earth, to expose the general system of that knowledge to the men with whom we live and to transmit it to the men who come after us so that the works of the past centuries are not useless for the centuries to come" (Diderot, 1994, p. 363). That is why, in addition to the sciences, the arts and the crafts, the *Encyclopédie* extends to the balance of the practical activities of the human genius, shows the virtues of work in its various dimensions, revalues the manual activity of the artisan giving it a dignity equivalent to that of the wise or the poet.[9]

But – it is also fair to say – if science has served the city, in turn, the city has sometimes served the science. Aware of how much it was beneficiary of scientific discoveries, the city has now and then protected the men of science (see the patronage of Galileo by the Florentine family of the Medici), has occasionally provided places for their work (see the *Uranienborg* observatory constructed by Frederic the II to Tycho Brahe). The city has even accepted to finance scientific activity without imposing any constraint to the liberty of research. A marvelous example is the reform of German university undertaken by Humboldt, in 1810, under three great pillars: complete liberty of research, full independence face to the state which

9. For further information on this practical dimension of the *Encyclopedie* and its iconic translation in the numerous engravings of its pages, cf. O. Pombo (2006).

ought to protect the autonomy of science and to pay for its activity and unrestricted cooperation among all those who are devoted to the service of truth.[10]

However, today, this disinterested relationship is no longer recognizable. Even if science and democracy were united by an intimate, constitutive and balanced relationship in the classical Greek world, and even if such reciprocity was maintained in certain periods and in some specific contexts of European History, the fact is that, today, we witness a much diverse state of affairs.

The situation begun to change in the beginning of the XX century. Science became more and more submitted to economic and political interests. And, in doing so, science seems not seeking anymore to understand the world and to search for the benefits it could provide to the city, as Bacon stressed. It seems to look for simply producing effects able to be useful to a private, minority sector of society, or to some military strategic plans. In a word, science has lost its original *ethos,* as Merton proposed to say,[11] and appears to have become nothing but an interested, pragmatic endeavour.

It is interesting to recover, even if in a much brief way, some illustrative moments of that process. In the very beginning of the XX century, the industrial and military scientific research structures which are being developed in Europe and in the USA are already visible. Unconnected with the university, those structures are directly financed by the economic power. See the case of the *Rockfeller Institute for Medical Reseach*, created in 1901, or the *Carnige Institute* of Washington, funded in 1902. In 1938, in the celebrated Szilard Appeal, scientific community accepted as legitimate not to publish the research results in areas connected with state defence plans. At the London conference of 1941, put forward by the *British Association for the Advancement of Science* in order to prepare the post-war, it was established that science ought to become the "effective laboratory of the best minds against war, poverty, all forms of discrimination, economic and political anarchy, and for the defence of human dignity, European awareness and world citizenship" (Sá

10. For the first time in the history of university, Humboldt's reform of Berlin University was rooted in the contributions – explicitly done for that purpose – by some of the most remarkable spirits of the time, namely Hegel, Fichte, Schleiermacher e Schelling. Cf. the volume edited by Jean-Luc Luc Ferry and Alain Renault (1979) where those foundational texts by Hegel, Fichte, Schleiermacher, Schelling and Humboldt are transcribed.

11. The term was coined by the American sociologist Robert K. Merton (1942) corresponding to the norms that should determine the *ethos* of scientific community, namely, universalism, communalism, disinterest and scepticism. A set of four norms to which was later added a fifth principle (Originality) thus giving rise CUDOS, the classical acronym used to designate the normative principles of true science. It is yet to be noted that these norms are close to the five novelties we have above signalized in Greek science (equality, refusal of authority, collectiveness/universality, dialectics, freedom of thought).

da Costa and Rémy Freire, 1943, p. 127). Almost at the same time, in September 1942, it was put forward the *Los Alamos Laboratory*, in *New Mexico*, directed by general Leslie Groves where, as well know, was developed the *Manhatan Project,* coordinated by Oppenheimer, in which were produced the three atomic bombs of 1945 of Trinity, Hiroshima and Nagasaki.

After the war, Alvin Weinberg (1961) and Solla Price (1963) identified what they named as "Big Science", the giant phenomenon which transformed scientific community in a set of employees, not anymore old *amateurs* and respected professors, but professionals who become obliged to respect a strict timetable, to fight for contracts, to design huge projects, to advertise their labs, to administrate large funds, etc. As a last example, in 1977, during the so called "cold war", circa of 1 million scientists worked in military projects and the third part of world scientific research was dedicated to the production of new weapons (Bridger, 2015). That is, during the XX century, decisive changings in the relation between science and political, economic and military power took place. Scientific community was obliged to negotiate its freedom and autonomy offering industrial applications, profit, prestige, military power in exchange of financial backing, grants, subsidies. By its side, political and economic power were able to more and more dominate, control and direct scientific activity. And that was done – we must not forget – in favour of their exclusive interest and profit, not in the interests of the social body in its all. As Jean Hamburger wrote, yet in the 90's of the past century, the political and economic power attitude towards science was not anymore, the (Humboldtian) "*laisser-faire*" and massive support, but instead the "*faire faire*" (Hamburger, 1991, p. 8). By other words, Humboldt become, each day more and more, a mirage.[12] Between science and university, which for Humboldt were the two faces of a same coin, it emerged the technological civilization and the competitive and market economy.

For its part, the philosophy of science prior to second world war tended to consider this situation as resulting from the *corruption* of the *Idea of Science*. Science is believed to be a clean cognitive endeavour, a pure theoretical activity away from any kind of practical objectives or technical purposes. That is the case of Husserl for whom the *Crisis of European Sciences* comes from the reduction of knowledge to material interests, from the technological domination of Nature

12. For Humboldt, University must be free of any exterior constraint and fully dedicated the search of truth. It ought to defend an autonomous, politically neutral science, freed from the tutelage of the church, freed from the authority and interests of the state, immune to pressure from bourgeois civil society interested in the usefulness of its results. The state was responsible for protecting the autonomy of university, guaranteeing its freedom and paying for science. On the reform of the German university, cf. Ferry, Pesron and Renault (1979).

and from the science's disregard towards the pursuit of the universal, ideal truth. On the contrary, for Husserl, what was important to safeguard was a contemplative, purely theoretical attitude. For Husserl, the theoretical attitude, based on a deliberate *epoche,* is fully unpractical, far from all applicatory, pragmatic interests (Husserl, 1936). In a parallel way, for Logical Positivism too, science is considered in its ideality, as a theoretical undertaking without any contact with the material conditions of its production. The meaning of a scientific statement comes exclusively from their internal linguistic components without any concern with the individual or the social conditions of its enunciation. As Carnap states, philosophy of science must take as its object the scientific statements in abstraction of the psychological and sociological conditions of its enunciation (Carnap, 1938, p. 43).

After Hiroshima and the awareness of the perverse effects of scientific activity over men's life and over world's equilibrium that was expanded since the second half of the XX century, philosophy of science came to recognize that the situation, namely the reduction of science to technoscience,[13] was a result of the very interested nature of science itself or, at least, of its actual condition. That is the case of Heidegger (1962), according to whom technoscientific developments are not the effect of a degenerated science. On the contrary, they are the extreme chapter of the Platonic and Aristotelian way of approaching Nature. Like Heidegger, Habermas (1968) recognises in technology, not a form of degeneration, but the very essence of knowledge which, in all its forms, is always oriented by some kind of interests.[14] And Lyotard (1979) stressed further that technoscience is neither the destiny of the scientific search for truth (Heidegger), nor the expression of an interested reason (Habermas), but the actual condition of human knowledge. His sceptical conclusion is that one of the main features of our post-modernity is the fact that science does not anymore intends to construct a theoretical description of the world but is reduced to the production of good performances, so to say, to a technological endeavour. As Lyotard writes: "What matters today is not the truth but the performance" (Lyotard, 1979: 91).

Now, the question is: Is it true that after Hiroshima, the traditional image of science and of its autonomy face to the political powers cannot be anymore

13. It is interesting to notice that the new term invented to designate this state of affairs – technoscience – instead of stressing the dependence of technique from scientific knowledge, points to the inverse situation: the dependence of science from technology. The dependence is so deep that the radical *logos* (still present in the word technology) disappears. (cf. Pombo, 2013, p. 57).

14. Behind hermeneutical interest (which constitutes the ground of the human sciences) and behind the emancipatory interest (which constitutes the root of philosophy), Habermas points out the technological interest (the one guiding the natural sciences), where knowledge is directed for the domination of Nature (Habermas, 1968).

maintained? Are we in fact condemned to the political promiscuity between the scientist and the sovereign? Between the scientist and the general? Between the scientist and the businessmen, that small parcel of the human community who, indeed, explores science for its own benefit and is greatly responsible by the perverse effects of science which we all regret? Or, even if often reduced to a merely operatory rationality submitted to the economic values and the strategies of domination of the planet, is it yet science, in the profundity of its equations, in the secret of its institutions, a pure and free activity?

Is it true that science does not anymore aim at giving meaning to the natural phenomena but just looks to produce operational, practical results able to answer private interests? Do we really face a radical alteration in the functioning regime of the very science? Or, what exists is just a change in the appropriation by political and economic powers of scientific results?

Is it true that science, which has always been instrumentalized by economic and political powers, today, it just continues to be so, but in a deepest way? Or, on the contrary, even if today scientific activity needs huge, gigantic financings, science is not at the service of any economic or political interests? Is it not that science, although its applications produce vast richness and its research activity requires today enormous amounts of funding, science as such is not – nor has ever been – at the service of any economic interests?

3. Now, I believe that the situation described above does not entirely corresponds to our present. In the last three decades, in addition to the interlocutors of the classic triangle constituted by the university, the state and the industry that prevailed during the XIX and XX century, science started to respond to new challenges, this time, not so much coming from political and economic power, but from civil society. We refer to a series of signs, each time more significant, that make us realize that science is today under the fire of a new public, more critical and intervening. A public that asks for accounts, sets conditions, demands compliance with certain rules, discusses the results and effects of scientific activity, that confronts science with new problems. As Isabel Stengers says, science today is faced with problems that it did not pose, but which are raised, or at least signalized, by the development of new public competences, "groups of citizens able to ask questions to which their interests make them sensitive, to demand explanations, to put conditions, to suggest modalities, to participate in the invention" (Stengers, 1993, p. 180).

Public opinion, which began to take shape in the salons, cafes and gazettes of the eighteenth century in France and in clubs and reading societies in England and

Germany,[15] is today an active interlocutor of science, a positive element that raises problems, determines objects of study, validates analyzes, supports research. It forces science to return to the concrete problems from which science had moved away. It forces science to look for integrated (interdisciplinary) solutions to the holistic issues it proposes. So are the countless pressure groups made up of citizens of different types, residents of a certain area, hemophiliacs, groups of students, pacifists, ecologists, environmentalists, etc. Concentrating large capacities of protest, these groups – which, according to Deleuze, should not be seen as new social classes but simply as "minorities" characterized by their ability to produce events, even if ephemeral, and by their "rebellious spontaneity" (Deleuze, 1990, p. 238) – are interested in the devastation of forests, hunger, energy, demographic explosion, environmental disasters. And they are strong enough to take the initiative of commenting, claiming, protesting on issues of safety, pollution, protection of species, nuclear winter, aids, atmospheric regime, ozone layer, in short, they have the capacity to jeopardize the mechanisms and results of science.

In other words, science is no longer dependent exclusively on political and economic powers, but also on the capacity of interrogation and on the critical intervention that the citizens and the public opinion hold today. A capacity which, as well known, is largely enhanced by the new mass media and digital technologies. Beyond an *authoritarian* science, in which scientists "know" and the public "does not know" and, therefore, cannot participate (and is not allowed to participate), what is currently underway is a participatory regime of science in which a new public, better informed, better prepared, better organized is willing to participate (and is welcomed to do so). What is currently in progress is a new science that accepts the alteration of the relations between those who ask the questions and those who answer them. A science that needs, and wants and requests an informed public, an interested public, a cultivated public able to participate, to criticize, to resist, to protest, in a word, a public able to exercise the healthy principle of civil surveillance.

4. However – and this is a key point of my argument – the democratic regime of science entails, both the development of the capacity of intervention and participation of citizens in the critical definition of science, and the reinforcement of the universal institutions that, as stressed above, constitute the necessary (transcendental) conditions for the autonomous advancement of scientific knowledge in its endless journey towards the truth.

15. On this subject, see the relevant work by Jurgen Habermas (1962) "Structural Change of the Public Sphere", pp. 46–93.

Only the effective and vigorous functioning of those universal configurations – the *scientific community*, the *school* (the *university*), the *library* and the *encyclopedia* – may counteract the effects of the domination over science by the authoritarian so-called "protection" or so-called "monetary support" of the political, economic, military powers. Only the healthy and accurate working of those cognitive institutions may guarantee an unbiased, self-directed, progressive scientific activity, guided by a genuine search for the truth.

We know that those universal institutions, have suffered a frightening erosion. As a result of the incomprehension of what constitutes their elevated destiny, each of them has been overwhelmed with the most diverse and strange tasks and functions. *School* has served to take care of the children while parents go to work. *Universities* claim to be professional instances, intended above all to hierarchically distribute the individuals in the labor market. *Library* can serve today to read the daily newspaper. Belonging to *scientific community* can be a means of gaining prestige and earning at the end of the month. And the *encyclopedia* (that is, the *internet)* can be used just to buy a domestic utensil, to know the time of a train or simply to send an email to a friend.

Nevertheless, however disfigured they may be, however violented they have been in their deepest purpose, they continue to serve science, its continuity and its democratic nature. And we all continue to understand that. We all know that, without them, science would be impossible. That is why we keep stressing that *school* must contribute to the formation of active citizens – the more knowledgeable the citizens are, the better the democratic process and the scientific activity will work. That is why *university*, even if submitted to quantity of pressures, continues to be a source of hope for scientific knowledge. That is why we keep fighting for the enrichment and actualisation of our *libraries*. Last but not least, that is why we vigorously defend the free, universal spread of internet (the *encyclopaedia* of today) as a force for the democratic transformation of science and for the relevant development of research: it enables the admission, the scope, the choice of scientific training, teaching, guidance to a large number of students independently of where they live and how much money they have (*school and university*), it allows easy, actualized and each day broader access to scientific bibliography (*library)*, it facilitates communication within *scientific community* and between the scientists and public.

Science has been (and continues to be) instrumentalized by exterior armed forces and external superpowers. But science did (does) not die. Not even the relativistic theory of double truth, the explosion of fake news, has succeeded to make it disappear men's desire of knowledge, men's will of truth. Scientific *ethos* has been severely threatened but, even though it has capitulated in many cases, it has not disappeared. It continues to live in the humble actions and functions and activities of the *librarians*, in the daily practices and technical procedures of the members of the *scientific community*, in the hard work and enthusiastic efforts of the *school masters and university professor*s as well as in the innovative performances of those young informaticians and engineers who, each day, produce the spectacular improvements of internet, that is, the *encyclopaedia* of our time.

Hence, in the perspective of a certain philosophy of science that I dare to defend here, what matters today – here and now – is to contribute for the reconstitution of those (transcendental) conditions of possibility of science, to invest in their restructuring, to reinforce their cognitive destiny and their capacity for contradicting the effects of the extrinsic domination of science, that is, for defending a democratic science. What is at stake is therefore a renewed attention to those universal institutions that are the library, the republic of the wise, the school and the encyclopedia. To be attentive to their current developments, to the trivial (sometimes bizarre) functions and tasks that, today, may be attributed to them, to the challenges to which they are faced. To recognize in them the shadow, both material and universal, both necessary and temporary, of science and democracy.

To finish, let me add that it is urgent to question of whether science, which was born in close conjunction with the democracy invented in Greece, which continues to have an internal democratic structure because of its ability to discuss its hypotheses, to test its results, to measure its own arguments and those of others, to establish discussions and controversies animated by the will of truth, and to accept the participation of public opinion and critical citizens, whether or not science is able to expand and reinforce its constitutive democratic nature. That can only be achieved by reinforcing the universal institutions that constitute the transcendental condition of possibility of both its cognitive development towards truth and its autonomy face to the political, military and economic interests, and, simultaneously, by opening the range of its interlocutors, by accepting the intervention of collective, interdisciplinary inventiveness in what has always been its own (private, specialized, cloistered) "domain" of work.

We rediscover here the issue of interdisciplinarity. Not so much in its cognitive dimension (as a decisive phenomenon of contemporary science which, since the middle of the XX century, is entering a process of transversal rationality, a process that entails sensitivity to complexity, ability to look for common mechanisms, attention to deep structures that can articulate what apparently cannot be articulated) but also as a open attitude of curiosity, of enjoying collaboration, cooperation, work in common. Without real interest in what the other has to say, interdisciplinarity is not possible. Interdisciplinarity implies to share our small domain of knowledge, to have the courage of abandoning the comfort of our technical language and to venture into a domain that belongs to everyone and of which nobody is the owner.

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